INCH-POUND

MIL-S-19500/509B
25 June 1993
SUPERSEDING
MIL-S-19500/509A
27 January 1989

The documentation and process conversion measures necessary to comply with this revision shall be completed by 26 Nov 93.

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPES 2N6338 AND 2N6341
JANS, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon power transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

Types	P_T 1/ T_A = +25°C	P_T T_C = +100°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_{STG} and T_{OP}
	W	W	V dc	V dc	V dc	A dc	A dc	°C
2N6338	200	112	120	100	6.0	25	10	-65 to +200
2N6341	200	112	180	150	6.0	25	10	-65 to +200

1/ Between T_C = +25°C and T_C = +200°C, linear derating factor (average) = 1.14 W/°C.

1.4 Primary electrical characteristics at T_A = 25°C.

Limit	h_{FE1}		$V_{BE(SAT)}$	$V_{CE(SAT)}$		C_{obo}
	I_C = 25 A dc, V_{CE} = 2.0 V dc	I_C = 10 A dc, V_{CE} = 2.0 V dc	I_C = 10 A dc, I_B = 1.0 V dc	I_C = 25 A dc, I_B = 2.5 A dc	I_C = 10 A dc, I_B = 1.0 A dc	1 MHz ≤ f ≤ 1 MHz, V_{CB} = 10 V dc, I_E = 0
			V dc	V dc	V dc	pF
Minimum	12	30	---	---	---	---
Maximum	--	120	1.8	1.8	1.0	450

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Research Laboratory, ATTN: AMSRL-EP-RD, Fort Monmouth, NJ 07703-5601, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics at T_A = 25°C - Continued.

Limit	$\|h_{FE}\|$	Pulse response		$R_{\theta JC}$
	f = 10 MHz I_C = 1.0 A dc V_{CE} = 10 V dc	t_{on}	t_{off}	
		µs	µs	°C/W
Minimum	4	---	---	----
Maximum	12	0.5	1.25	.875

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 11, T-4A, and herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein.

3.4 Marking. Devices shall be marked in accordance with MIL-S-19500. At the option of the manufacturer, the marking of the country of origin may be omitted from the body of the transistor.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CEX1} and h_{FE1}	Not applicable
11	I_{CEX1} = 100% of initial value or 3 μA dc, whichever is greater; Δh_{fe} = ±25% of initial value.	I_{CEX1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX1} = 100% of initial value or 3 μA dc, whichever is greater; Δh_{FE2} = ±25% of initial value.	Subgroup 2 of table I herein; ΔI_{CEX1} = 100% of initial value or 3 μA dc, whichever is greater; Δh_{FE2} = ±25% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

V_{CB} = 20 V dc minimum

T_J = +187.5°C ±12.5°C

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX), of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B3	2037	Test condition A, all internal leads for each device shall be pulled separately.
B4	1027	V_{CE} = 20 V dc; ΔT_J between cycles ≥ 100°C; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	2037	Test condition A, all internal leads for each device shall be pulled separately.
B3	1027	V_{CE} = 20 V dc; ΔT_J between cycles ≥ 100°C; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
C2	2036	Test condition A; weight = 10lbs.; time = 15 s.
C6	1026	V_{CE} = 20 V dc; ΔT_J between cycles ≥ 100°C; $t_{on} = t_{off}$ = 3 minutes minimum for 6,000 cycles.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurement shall be conducted in accordance with test method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be 3.0 A dc.

b. Collector to emitter voltage magnitude shall be 10 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be 25°C ≤ t_r ≤ 75°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to header.

f. Maximum limit for $R_{\theta JC}$ shall be .875°C/W.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Bias Condition D, I_C = 50 mA dc pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N6338				100		V dc
2N6341				150		V dc
Collector to emitter cutoff current	3041	Bias Condition D,	I_{CEO}		50	µA dc
2N6338		V_{CE} = 50 V dc				
2N6341		V_{CE} = 75 V dc				
Emitter to base cutoff current	3061	Bias Condition D, V_{EB} = 6 V dc	I_{EBO}		100	µA dc
Collector to emitter cutoff current	3041	Bias Condition A, V_{BE} = -1.5 V dc	I_{CEX1}			
2N6338		V_{CE} = 100 V dc			10	µA dc
2N6341		V_{CE} = 150 V dc			10	µA dc
Collector to base cutoff current	3036	Bias Condition D	I_{CBO}			
2N6338		V_{CB} = 120 V dc			10	µA dc
2N6341		V_{CB} = 180 V dc			10	µA dc
Base to emitter saturation voltage	3066	Test condition A; I_B = 1.0 A dc; I_C = 10 A dc; pulsed (see 4.5.1)	$V_{BE(SAT)}$		1.8	V dc
Collector to emitter saturation voltage	3071	I_B = 1.0 A dc; I_C = 10 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)1}$		1.0	V dc
Collector to emitter saturation voltage	3071	I_B = 2.5 A dc; I_C = 25 A dc; pulsed (see 4.5.1)	$V_{CE(SAT)2}$		1.8	V dc
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 0.5 A dc; pulsed (see 4.5.1)	h_{FE1}	40		
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE2}	30	120	

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Con't						
Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 25 A dc; pulsed (see 4.5.1)	h_{FE3}	12		
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias Condition A, V_{BE} = -1.5 V dc	I_{CEX2}			
2N6338		V_{CE} = 100 V dc			1.0	mA dc
2N6341		V_{CE} = 150 V dc			1.0	mA dc
Low-temperature operation		T_A = -55°C 2/				
Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 10 A dc; pulsed (see 4.5.1)	h_{FE4}	10		
Subgroup 4						
Pulse response:	3251	Test condition A except test circuit and pulse requirements in accordance with figure 1.				
Turn-on time		V_{CC} = 80 V I_C = 10 A dc I_{B1} = 1.0 A dc	t_{on}		0.5	µs
Turn-off time		V_{CC} = 80 V I_C = 10 A dc I_{B1} = I_{B2} = 1.0 A dc	t_{off}		1.25	µs
Storage time		V_{CC} = 80 V I_C = 10 A dc I_{B1} = I_{B2} = 1.0 A dc	t_s		1.0	µs
Magnitude of common emitter, small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 1.0 A dc; f = 10 MHz	$\|h_{FE}\|$	4.0	12	
Open capacitance open circuit	3236	V_{CB} = 10 V dc; I_E = 0; 0.1 MHz ≤ f ≤ 1 MHz	C_{obo}		450	pF

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (dc operation)	3051	T_C = +25°C; t = 1 s; 1 cycle (see figure 2)				
Test 1						
(Both device types)		I_C = 25 A dc V_{CE} = 8 V dc				
Test 2						
(Both device types)		I_C = 14 A dc V_{CE} = 14 V dc				
Test 3						
2N6338		I_C = 100 mA dc V_{CE} = 100 V dc				
2N6341		I_C = 66 mA dc V_{CE} = 150 V dc				
Safe operating area (switching)	3053	Load condition C; (unclamped inductive load) see figure 3				
		T_C = +25°C; duty cycle ≤ 10 percent; R_s = 0.1Ω; t_r = t_f ≤ 500 ns				
Test 1		t_p = 5 ms (vary to obtain I_C); R_{B81} = 100Ω; V_{B81} = 20 V dc; R_{B82} = ∞; V_{B82} = 0; V_{CC} = 50 V dc; I_C = 20 A dc; L = 3 µH				
Test 2		t_p = 5 ms (vary to obtain I_C); R_{B81} = 100Ω; V_{B81} = 10 V dc; R_{B82} = ∞; V_{B82} = 0; V_{CC} = 50 V dc; I_C = 200 mA dc; L = 10 mH				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Con't						
Safe operating area (switching)	3053	Clamped inductive load; T_A = +25°C; duty cycle ≤ 5 percent; t_p = 1.5 ms (vary to obtain I_C); V_{CC} = 50 V dc; I_C = 25 A dc; (see figure 4)				
2N6338		Clamped voltage = 100 V dc				
2N6341		Clamped voltage = 150 V dc				
Electrical measurements		See table II, steps 1 and 5.				

1/ For sampling plan see MIL-S-19500.
2/ In this subgroup, the sample units subjected to the high-temperature operation test shall be permitted to return to and be stabilized at room ambient temperature prior to their being subjected to the low-temperature operation test.

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TABLE II. Groups B and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = -1.5 V dc	I_{CEX1}			
	2N6338		V_{CE} = 100 V dc			10	µA dc
	2N6341		V_{CE} = 150 V dc			10	µA dc
2	Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = -1.5 V dc	I_{CEX2}			
	2N6338		V_{CE} = 100 V dc			1.0	mA dc
	2N6341		V_{CE} = 150 V dc			1.0	mA dc
3	Collector to emitter voltage (saturated)	3071	I_C = 25 A dc; I_B = 2.5 A dc pulsed (see 4.5.1)	$V_{CE(sat)1}$		1.8	V dc
4	Base to emitter voltage (saturated)	3066	Test condition A; I_C = 10 A dc; I_B = 1.0 mA dc pulsed (see 4.5.1)	$V_{BE(sat)}$		1.8	V dc
5	Forward-current transfer ratio	3076	V_{CE} = 2.0 V dc; I_C = 0.5 A dc; pulsed (see 4.5.1)	h_{FE1}	40		
6	Forward-current transfer ratio	3076	V_{CE} = 2 V dc; I_C = 10 A dc pulsed (see 4.5.1)	h_{FE2}	30	120	
7	Collector to emitter cutoff current	3041	See step 1	ΔI_{CEX1}	100 percent of initial value or 2 mA dc, whichever is greater.		
8	Forward-current transfer ratio	3076	See step 5	Δh_{FE1}	±25 percent change from previously measured value.		

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroups 3, 4, and 5, .see table II herein, steps 1, 5, 7, and 8.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 5.

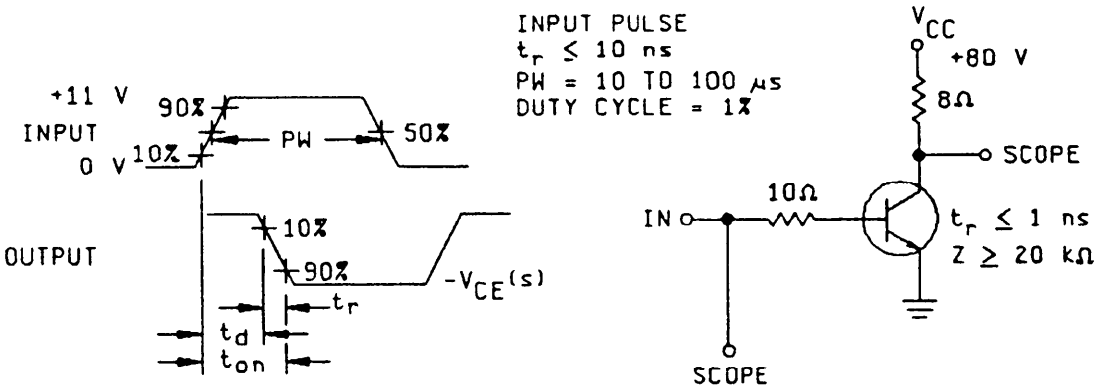
b. Subgroups 3 and 6, see table II herein, steps 2, 5, and 8.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

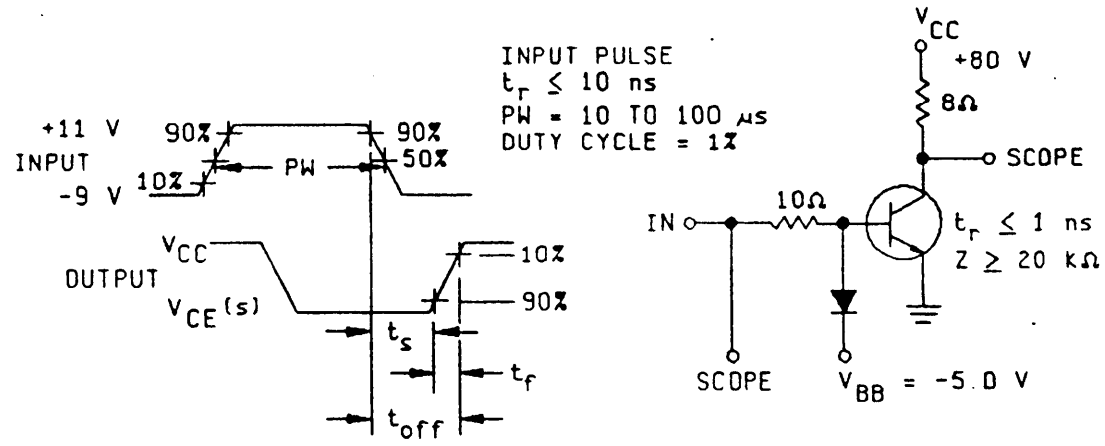
a. Subgroups 2 and 3, .see table II herein, steps 1 and 5 for all levels.

b. Subgroup 6. .see table II herein, steps 2, 5, and 8 for all levels.

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TURN-ON (t_{on}) TIME TEST CIRCUIT



TURN-OFF (t_{off}) TIME TEST CIRCUIT

FIGURE 1. Switching time test circuits.

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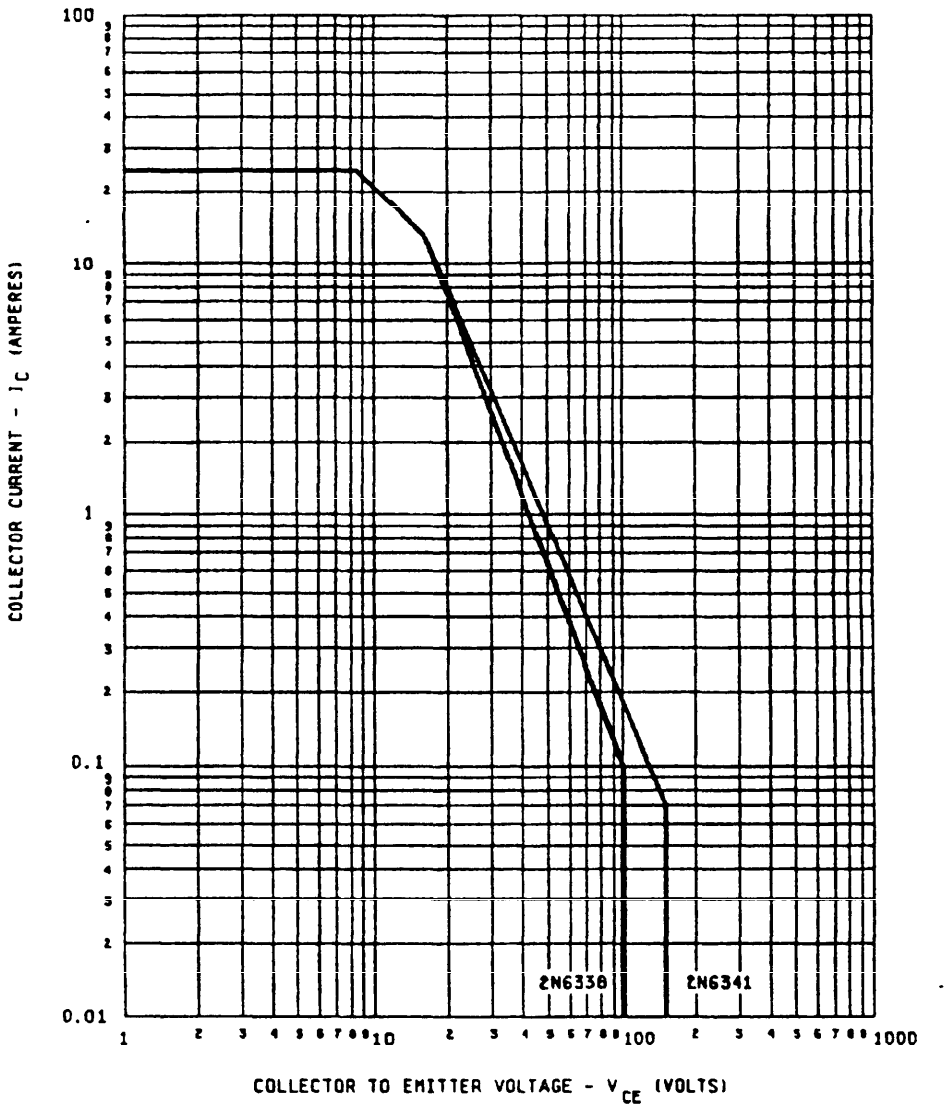


FIGURE 2. Maximum safe operating area graph (continuous dc) for types 2N6338 and 2N6341.

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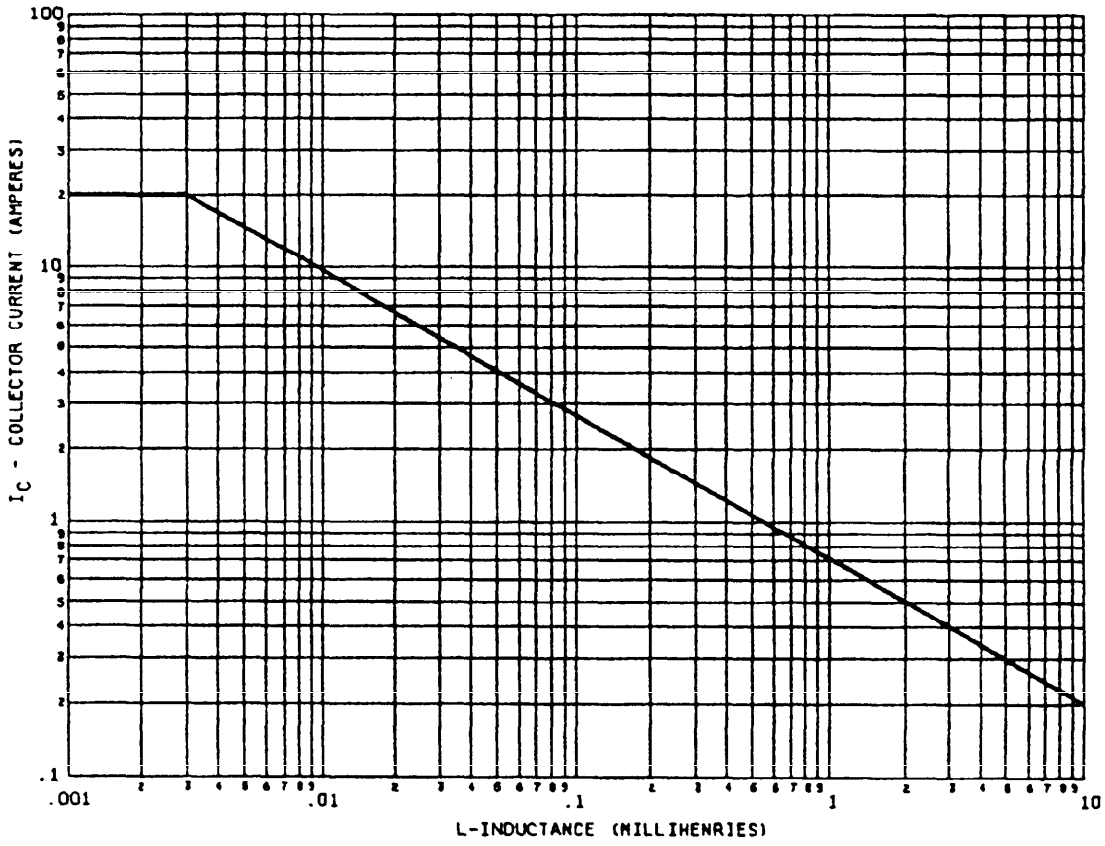
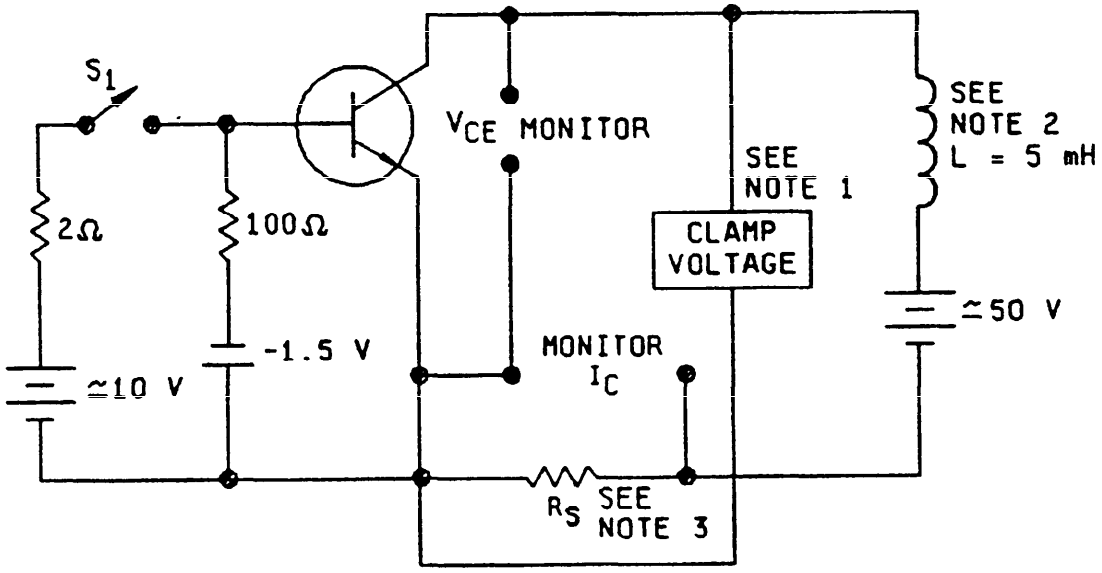


FIGURE 3. Safe operating area for switching between saturation and cutoff - unclamped inductive load.

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Procedure:

1. With switch S_1 closed, set the specified test conditions.
2. Open S_1. Device fails if clamp voltage not reached and maintained until the current returns to zero.
3. Perform specified end-point tests.

NOTES:

1. Either a clamping circuit or clamping diode may be used.
2. The coil used shall provide a minimum inductance of 5 mH at 25 A with a maximum dc resistance of .1Ω. For reference only: 4 Triad C-48U; (20 mH windings in parallel) or equivalent.
3. $R_S \leq .1\Omega$, 12 W, 1 percent tolerance maximum (noninductive).

FIGURE 4. Clamped inductive sweep test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Design and application guidance. The following PNP type transistors are complementary to the NPN devices listed herein.

Transistor (NPN)	Complementary (PNP) transistor types
2N6338	2N6437
2N6341	2N6438

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Air Force - 17
NASA - NA

Review activities:
Air Force - 85, 99
DLA - ES

User activity:
Air Force - 80

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1430)

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